

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304

Course Title: AI Assisted Coding

Year/Sem: III / II

Regulation: R23

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Week: 1 – Wednesday

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Batch: 46

Lab Objectives

- Install and configure GitHub Copilot in VS Code
- Explore AI-assisted code generation
- Analyze correctness and efficiency of AI-generated code
- Understand prompt-based coding workflow

Task 1: Fibonacci without Functions

Python Code

```
n = int(input("Enter number: "))
```

```
a, b = 0, 1
```

```
for i in range(n):  
    print(a, end=" ")  
    a, b = b, a + b
```

Explanation: Fibonacci sequence is generated using a loop without using functions.

Task 2: Optimized Fibonacci Code

Python Code

```
n = int(input("Enter number: "))

a, b = 0, 1

for _ in range(n):
    print(a, end=" ")
    a, b = b, a + b
```

Explanation: Optimized by reducing unnecessary variables and improving readability.

Task 3: Fibonacci using Functions

Python Code

```
def fibonacci(n):
    a, b = 0, 1
    for i in range(n):
        print(a, end=" ")
        a, b = b, a + b

n = int(input("Enter number: "))
fibonacci(n)
```

Explanation: Logic is modularized using a function for better reuse and clarity.

Task 4: Comparison

- Procedural code is simple but less reusable
- Function-based code improves readability
- Functions are better for large applications

Task 5: Iterative vs Recursive Fibonacci

Iterative Code

```
def fib_iter(n):
    a, b = 0, 1
    for i in range(n):
        print(a, end=" ")
        a, b = b, a + b
```

Recursive Code

```
def fib_rec(n):
    if n <= 1:
        return n
    return fib_rec(n-1) + fib_rec(n-2)

for i in range(10):
    print(fib_rec(i), end=" ")
```

Explanation:

- Iterative approach is faster and memory efficient
- Recursive approach is simple but slower

Conclusion

GitHub Copilot helps in generating code quickly.

However, developers must verify correctness, optimize logic, and ensure readability for real-world applications.